ChatGPT

Hi

Hi. What do you need?

Actaully I am going to start data visualisation to start learn now can you help me out

Thought for 15s

Yes. Start properly, otherwise you will waste time making pretty charts without understanding anything.

Here is the right way to begin **data visualisation**.

What data visualisation actually is

Data visualisation is not "making graphs."
It is **turning data into something a human can understand quickly and correctly**.

A good visual should answer one of these:

- comparison
- trend over time
- distribution
- relationship
- composition

If a chart does not answer a clear question, it is useless.

Learn these 5 chart purposes first

1. Comparison

Use when you want to compare categories.
Examples:

- sales by product
- marks by subject

- number of customers by city

Best charts:

- bar chart
- column chart

2. Trend

Use when data changes over time.
Examples:

- monthly revenue
- daily website visits
- yearly temperature

Best charts:

- line chart

3. Distribution

Use when you want to know how values are spread.
Examples:

- student marks
- salaries
- ages

Best charts:

- histogram
- box plot

4. Relationship

Use when checking whether two variables move together.
Examples:

- study hours vs marks
- ad spend vs sales

Best charts:

- scatter plot

5. Composition

Use when showing parts of a whole.
Examples:

- budget split
- market share

Best charts:

- stacked bar chart
- pie chart only if categories are very few and simple

First things you must understand

Before creating charts, understand these terms:

- **variable**: a column in data
- **categorical variable**: names/groups like gender, city, product
- **numerical variable**: numbers like age, salary, marks
- **time variable**: date, month, year
- **dataset**: full table of rows and columns

If you do not know what type of variable you have, you will choose the wrong chart.

The beginner mistakes you should avoid

These are common and bad:

- using pie charts for everything
- adding too many colours
- 3D charts
- using charts before cleaning data
- making a chart without asking a question first
- putting too much text and decoration

A chart should be clear, not fancy.

Your learning order

Follow this order exactly:

Stage 1: Understand data types

Learn to identify:

- categorical
- numerical
- ordinal
- time-based

Stage 2: Match question to chart

Ask:

- am I comparing?
- showing trend?
- checking distribution?

- finding relationship?
- showing composition?

Stage 3: Learn the basic charts

Master these first:

- bar chart
- line chart
- histogram
- box plot
- scatter plot

Stage 4: Learn design principles

Understand:

- titles
- labels
- axis scale
- colour use
- sorting
- avoiding clutter

Stage 5: Interpret charts

This matters more than drawing them.
You should be able to say:

- what is highest
- what is lowest
- what trend exists
- whether data is skewed
- whether outliers exist
- whether variables are correlated

What you should do today

Do this in one sitting:

Task 1

Take a small dataset with:

- 1 categorical column
- 1 numerical column
- 1 date/time column if possible

Task 2

Create:

- one bar chart
- one line chart
- one histogram
- one scatter plot if there are two numeric columns

Task 3

For each chart, write 2 lines:

- what the chart shows
- one insight from it

That is how you learn. Not by only watching videos.

Example thinking process

Suppose your data has:

- `Month`
- `Sales`
- `Product Category`

Then:

- compare categories → bar chart
- monthly change in sales → line chart
- distribution of sales values → histogram

That is the mindset you need.

Best tools to start with

Choose one based on your course:

- **Excel**: easiest for basics
- **Tableau**: best for visual analytics and dashboards
- **R / ggplot2**: best for academic learning and deeper control
- **Python / matplotlib / seaborn**: best for programming-based analysis

Since you are studying data-related subjects, **Tableau + R or Python** is a strong combination.

A 7-day beginner plan

Day 1

Understand data types and chart purposes

Day 2

Learn bar chart and line chart

Day 3

Learn histogram and box plot

Day 4

Learn scatter plot and correlation basics

Day 5

Learn bad vs good chart design

Day 6

Take one dataset and make 5 charts

Day 7

Write insights from each chart and explain them in plain English

What “good progress” looks like

After one week, you should be able to:

- choose the correct chart without guessing
- explain why that chart is correct
- create basic visuals
- write simple insights from them

If you cannot explain the insight, you do not understand the chart yet.

My direct advice

Do not start with dashboards.
Do not start with advanced tools.
Do not memorise chart names blindly.

Start with:

1. question
2. variable type
3. chart selection
4. interpretation

That is the core.

I can take you through this in a proper way from scratch and teach you lesson by lesson.
We can start right now with **Lesson 1: data types and choosing the correct chart**.

so my lab session is having tableau in it i do have used excel but dont have the experience in data cleaning and pre processing

Thought for 9s

That is not a big problem. For **beginning Tableau**, you do **not** need deep preprocessing knowledge on day one. But you do need to understand the **basics of clean data**, otherwise your charts will be wrong.

What matters first is this:

What Tableau expects

Tableau works best when your data is:

- in rows and columns
- one row = one record
- column names are clear
- no random merged cells
- no empty rows on top
- no totals/subtotals inside the raw data
- dates stored as dates
- numbers stored as numbers
- categories spelled consistently

If your Excel file is messy, Tableau will not fix your thinking for you.

What you must learn first in data cleaning

You only need these beginner-level preprocessing concepts first:

1. Remove structural mess

Bad raw files usually have:

- merged cells
- multiple headers
- blank rows
- extra notes above the table
- total rows inside the dataset

These should be removed before analysis.

2. Fix inconsistent values

Example:

- `Male` , `male` , `M`
- `London` , `london`

These must be standardized, or Tableau treats them as different categories.

3. Check missing values

You must ask:

- are blanks acceptable?
- should I remove those rows?
- should I replace them?

For now, just learn to **identify** missing values.

4. Correct data types

A column should be clearly one of these:

- text
- number
- date
- boolean/geographic if needed

If sales is stored as text, your analysis becomes garbage.

5. Remove duplicates

If the same record appears twice, your counts and sums will be misleading.

For Tableau specifically

In most beginner lab sessions, you will usually do:

1. connect to Excel or CSV
2. inspect fields
3. correct data types
4. rename columns
5. filter unnecessary records
6. make charts

So your preprocessing at this stage is usually **light cleaning**, not hardcore data engineering.

What you should know in Tableau from the start

Learn these first:

- **Dimensions** = categorical fields like city, product, gender
- **Measures** = numeric fields like sales, profit, marks
- **Rows and Columns shelves**
- **Filters**
- **Marks card**
- **Show Me**
- **Sorting**
- **Date hierarchy**

If you understand dimensions vs measures properly, half the confusion disappears.

Your practical learning order

Do it in this order:

Step 1: Learn what clean data looks like

Take one Excel sheet and inspect:

- column names
- blank cells
- duplicates
- wrong spellings
- wrong date formats

Step 2: Load it into Tableau

When Tableau reads the data:

- check whether each field type is correct
- change text/number/date where needed

Step 3: Build only 4 charts first

Start with:

- bar chart
- line chart
- histogram
- scatter plot

Do not jump to dashboards yet.

Step 4: Write insight from each chart

Example:

- which category is highest?

- is there an upward trend?
- are there outliers?
- do two variables move together?

What you do not need right now

Do not worry yet about:

- feature engineering
- normalization
- standardization
- one-hot encoding
- advanced imputation
- ETL pipelines

Those matter more in machine learning and advanced analytics, not in your first Tableau learning phase.

Your mindset should be this

Before making a chart, ask:

1. Is my data clean enough?
2. What is each column type?
3. What question am I answering?
4. Which chart fits that question?

That is the correct workflow.

A small example

Suppose your Excel file has:

Student	Subject	Marks	Exam Date
A	Math	78	01/02/2026
B	Science	85	03/02/2026

Then in Tableau:

- `Student` , `Subject` → dimensions
- `Marks` → measure
- `Exam Date` → date field

You can build:

- bar chart of marks by subject
- line chart if you have marks over time

- histogram of marks distribution

What you should do in your next lab

Before touching Tableau, spend 10 minutes checking the file:

- are headers clean?
- any blanks?
- any duplicates?
- any totals in the table?
- are dates and numbers proper?

Then in Tableau:

- connect data
- verify data types
- build one bar chart and one line chart
- explain the result in plain English

Brutal truth

A lot of beginners think they are weak at visualisation, but the real issue is they cannot **read the dataset properly**.
If you can identify columns, types, errors, and the right chart, you are already on the correct path.

The gap is not "I don't know preprocessing."
The gap is usually "I don't yet know what to check before visualising."

We can fix that fast.

Send me either:

- a sample dataset structure, or
- the topic from your Tableau lab,

and I will break it into exact steps you should do in class.

1-Introduction to Data …
Presentation

2-Encoding for percepti…
Presentation

3-Representation.pptx
Presentation

4-Presentation.pptx
Presentation

5-Interaction.pptx

Week 24 (6) - Dashboard…

Presentation Presentation

sure now ill upload the lecture slides if past 6 weeks you just analyse them and teach me the important ones and yet to known before landing into lab also make sure if any important details are there like any announcement or anything kindly inform me

I'm pulling out two things separately: the concepts you actually need for lab survival, and the admin details that can affect coursework or deadlines. After that I'll condense the six weeks into a usable study path.

I've got the admin pieces. One thing is already clear: the module is not teaching Tableau as a button-click exercise. It's teaching why a view works or fails, and that will matter in labs and in the report.

Thought for 2m 2s

I went through the six uploaded decks. Here is the version that actually matters before you step into lab.

What this module is really teaching

This module is **not mainly about Tableau buttons**. It is about learning to **ask the right question, choose the right chart, encode it correctly, present it cleanly, and add only useful interaction**. Tableau is just the implementation tool. That is obvious from the module aim, learning outcomes, and the repeated focus on perception, chart choice, dashboard design, interactivity, and visual storytelling. 1-Introduction to Data Visualis…

The six-week idea flow

The lectures build in a very clean order:

Week 18 — Introduction: what data visualisation is, why it helps cognition, the main relationship types you can show, and common ways visuals fail or mislead. The core message is that visuals help people explore, understand, and communicate data, largely because they reduce working-memory burden and make patterns visible.

Week 19 — Encoding for perception: how humans actually read visuals. This is where pre-attentive perception, Gestalt grouping, Bertin, Cleveland & McGill, and encoding quality come in. This is foundational. If you ignore this, your charts may still "work" technically, but they become harder to read or misleading.

Week 20 — Representation: how to choose the right chart for the question. This is the most directly useful lecture for lab work because it links data type + analytical task + chart type. It also introduces the Tableau language of **Dimensions**, **Measures**, **Discrete**, and **Continuous**.

Week 21 — Presentation: once you choose the right chart, you still have to make it readable. This lecture is about form following function, reducing chartjunk, data-ink ratio, effective use of bars/lines/points, legends, scales, and arranging multiple views.

Week 22 — Interaction: how a static chart becomes an exploratory tool. This covers selection, tooltips, sorting, filtering, parameters, local/global filters, brushing and linking, zooming, and how interaction affects coursework grading for complex questions.

Week 24 — Dashboards and Paper Landscapes: how to turn all of the above into a single-screen dashboard and how to document your design concept before building it. This is where the coursework becomes concrete.

What you absolutely need to know before lab

If you remember only a few things, remember these.

1. Always start with the question

Do **not** start with "let me make a chart."
Start with: **what am I trying to find out?**

The slides repeatedly frame chart choice around analytical relationships such as nominal comparison, time series, ranking, part-to-whole, deviation, distribution, and correlation. That is the logic your labs will expect.

2. Know your data types

You need to quickly identify:

- **quantitative** variables
- **categorical** variables
- categorical subtypes like **nominal**, **ordinal**, **interval bins**, and **hierarchical**

This matters because the wrong chart usually comes from not understanding the variable types. 3-Representation

3. In Tableau, know these four words cold

- **Dimensions** = usually categories
- **Measures** = usually numbers
- **Discrete** = blue

- **Continuous** = green

If you are confused in lab, first check whether Tableau classified the field correctly.

4. Position is the strongest encoding

For quantitative information, the slides make it very clear that **position** is the best visual encoding, followed by **size**. Once you start overloading charts with too many colours, shapes, or other retinal variables, reading becomes slower and less accurate.

5. Respect the three-variable limit

Bertin's logic in the lecture is simple: a chart works best when it stays close to **two planar variables** and at most **one retinal variable**. Go beyond that and the chart becomes effortful to decode. That is why overloaded dashboards and flashy charts usually fail.

6. Choose charts by task, not preference

The representation lecture is the practical core:

- **Bar chart** for comparison, ranking, deviation
- **Line chart** for time series
- **Histogram** for distribution
- **Scatterplot** for correlation, anomalies, clusters
- **Pie chart** only for simple part-to-whole with few categories and large differences
- **Treemap / choropleth / bubble / area charts** only when the task genuinely fits them

Also, the slides explicitly warn that stacked area charts are hard to read, especially for comparisons and absolute value retrieval.

7. Bars must start at zero

This is non-negotiable for bars. The presentation lecture treats bar bases seriously because truncating a bar axis distorts perception. If you need a non-zero scale, use another mark type.

8. Avoid chartjunk

You will lose marks in practice even if the chart is technically correct but visually noisy. The decks repeatedly warn against:

- 3D
- unnecessary colours
- decorative clutter
- over-strong gridlines
- misleading axes

- titles/labels that bias interpretation

Form follows function. Not the other way round.

9. Use interaction only when it helps answer a harder question

Simple questions should be answerable **without** interaction. Interaction matters mainly when answering a **complex** question. For better grades, the interaction should not be random — it should clearly help the user filter, coordinate, compare, or drill down.

10. A dashboard is not a collage

When multiple views are used, the slides want:

- logical grouping
- consistent scales and colour meaning
- natural reading order
- top-left prioritisation for key views
- no unnecessary legends or scrollbars
- clear proximity/similarity between related views

That is dashboard thinking. Not just placing charts on a screen.

The most important concepts from each lecture, in plain English

Week 18: why visualisation is powerful

The strongest takeaway is that visualisation is an **external thinking tool**. Humans have limited working memory, and visuals help by making relationships spatially obvious. The lecture also introduces the core analytic relationship types and warns early against misleading design choices like manipulated axes, 3D, and bad colour use.

Week 19: how the eye reads a chart

This week is about **pre-attentive perception** and **Gestalt principles**.

What matters for you:

- people notice **position, colour, form, motion** quickly
- grouping comes from **proximity, similarity, continuity, common fate, connection, enclosure**
- colour should be used carefully
- strong saturated colours should be for emphasis, not everything
- 3D perspective is usually a bad idea
- too many visual encodings make charts harder to read

This week explains why some charts “feel easy” and others feel messy.

Week 20: how to pick the chart

This is the lecture you should revise first before lab.

What matters:

- use a **table** when you need precise lookup
- use a **graph** when you need to see relationships
- identify the relationship first, then pick the chart
- Tableau's **Dimensions/Measures** and **Discrete/Continuous** are central

If you can classify the question as comparison / trend / distribution / correlation / part-to-whole / deviation, you can usually pick the correct visual.

Week 21: how to make the chart readable

This lecture is where many students lose quality.

Important rules:

- reduce non-data ink
- emphasise the important marks
- do not overload one chart
- horizontal bars are often better for ranking
- legends are not always needed; line charts often work better with direct labels
- keep multiple lines/subdivisions limited, ideally around 5–8
- use small multiples when one chart becomes overloaded

This is the lecture that stops you from making ugly-but-correct visuals.

Week 22: how interaction should work

This is directly relevant to Tableau.

Important interaction ideas:

- **selection** for focus
- **tooltips/details on demand**
- **sorting**
- **filter widgets**
- **parameters**
- **local vs global filters**
- **brushing and linking**
- **zoom/pan** mainly for maps
- interaction should have good feedback and tight coupling

For coursework, the slides explicitly say that a credible use of an interactive filter/parameter helps for **B**, while credible coordination of multiple views helps for **A** on the complex question.

Week 24: dashboards and proposal thinking

This week gives you the bridge from theory to coursework.

Key takeaways:

- your dashboard should answer questions efficiently, not just show charts
- place related views together
- shared filters can be more elegant than separate widgets everywhere
- remove scrollbars when possible
- use fixed axes when continuity matters across filtered views
- choose colours semantically, not randomly
- paper landscapes are for **pre-implementation design thinking**, not polished Tableau screenshots

The model dashboard discussion is useful because it shows concrete improvements rather than vague theory.

What seems most important for your first Tableau labs

Given what you told me earlier, this is the exact knowledge you need first:

1. **Field recognition**
   Look at each column and decide: dimension or measure? discrete or continuous? date or text or number?
2. **Question to chart mapping**
   Learn to say: “This is a ranking question, so bar chart.” Or “This is a trend question, so line chart.”
3. **Basic encoding discipline**
   Put the strongest variable on position. Use colour only if it genuinely separates categories or highlights something important.
4. **Simple interaction first**
   Learn filters, sorting, and brushing before worrying about fancy dashboards.
5. **Clean presentation**
   No 3D, no clutter, no weird palettes, no truncated bar axes.

Important admin details and announcements I found

These are the ones worth paying attention to.

The final assessed submission for **CS5803** is a **written report of up to 3000 words plus a Tableau solution**, due **5 May 2026 at 11:00 am**.

The proposal is described in the intro as **formative and optional**, but the later dashboard lab gives the concrete workflow: prepare a **two-page proposal** and email it to your marker on **15 April 2026** for verbal feedback in **week 30 on 20 April 2026**. The same week also says slots are limited and earlier submissions get priority.

The proposal should include a **paper landscape** of your design ideas, and the slides explicitly say **do not prototype the proposal in Tableau**. Sketch the views and interactions instead.

For the coursework dashboard, the end product should be **a single interactive dashboard fitting within 1920 × 1080**. The week 24 deck also advises using a **fixed size of 1920 × 1080 or lower**, because larger or automatic sizes may render badly on the marker's screen.

When sharing the coursework, the recommended format is a **packaged workbook (.twbx)**. The slides also explicitly warn: **do not publish to Tableau Public until grades are released**.

The next lab after week 24 was going to cover the **2-page proposal document** and the **Stories feature in Tableau** for the walkthrough section. Then **week 26** is a **Tableau Masterclass by Luke Stoughton** and is described as the **last regular lab session**.

Support resources mentioned in the slides:

- solution workbooks for previous lab activities are on **Brightspace**
- the **Third Party presentations** folder has design guidance
- Brightspace has dataset suggestions
- the instructor said previous full submissions would **not** be published, though extracts may be shown for guidance

There is also an older, clearly time-bound notice about **coursework groups**: if a group had been formed, students were asked to let the lecturer know **ASAP / by end of that day**. That looks like a past classroom announcement rather than an upcoming action for you now.

The practical bottom line

For your lab, the most important thing is not preprocessing theory. It is this workflow:

question → **data type** → **relationship** → **chart** → **encoding** → **presentation** → **interaction**

If you can do that, you are already on track.

The two lectures I would tell you to revise first are:

Week 20 Representation and **Week 21 Presentation**.

Then revise **Week 22 Interaction** because that is where Tableau dashboard logic becomes important.

What you should do next

Before your next lab, make sure you can answer these without hesitation:

- Is this field a dimension or a measure?
- Is this question comparison, trend, distribution, correlation, deviation, or part-to-whole?
- Which chart fits that task?
- Is my encoding overloaded?
- Is my view clean enough to read quickly?
- Would interaction actually help, or am I adding it just because Tableau allows it?

That is the standard you need.

Send me the **lab sheet** next, and I'll turn it into a ruthless step-by-step survival guide for the class.

Sources

6-Dashboards and Infog...
Presentation

Analysis and Design 202...
PDF

Visualisations for Machi...
PDF

understand these things

Thought for 36s

Yes. These three files add the **design side** and a **special topic side** to what you already uploaded. Here is what you need to understand properly.

1. `6-Dashboards and Infographics`

This lecture is telling you one blunt fact: **dashboard and infographic are not the same thing**.

A **dashboard** is for **monitoring**. It should show the most important information on a **single screen**, so someone can understand the status of something **at a glance**. It is **not** mainly for deep exploration, long reading, or detailed reporting. If it needs scrolling or jumping across screens, it has already failed as a dashboard. 6-Dashboards and Infographics

An **infographic** is for **communication/storytelling**. Its job is to explain something clearly and quickly, often by combining several views into a narrative. The lecture stresses that a visual should not be "beautiful" for decoration; it should be understandable first. That is the central rule. 6-Dashboards and Infographics

For your coursework, this matters because Tableau calls any composition of views a "dashboard," but the lecture explicitly says that **most students will probably end up making something closer to an infographic that tells a story**, with the questions acting as the structure of that story. 6-Dashboards and Infographics

The common dashboard mistakes are also important:

- too much on one screen
- not enough context
- excessive detail
- bad arrangement
- misleading or inaccurate quantitative encoding
- clutter and meaningless visual effects

So the real lesson is simple:
Do not build a collage of charts. Build a purpose-driven screen.

2. `Analysis and Design 2025-26`

This is probably one of the most important files for your coursework thinking, because it explains **how to design before touching Tableau**.

The lecture frames visualisation design like a development lifecycle:

- define and analyse requirements
- design
- implementation
- testing/evaluation
  And it says this should be done in a **user-centred, iterative** way.
  Analysis and Design 2025-26

The first serious idea here is **requirements analysis**. Before building anything, you are supposed to understand:

- the domain

- the users
- their goals, skills, and experience
- the device/context where the visual will be used Analysis and Design 2025-26

That is why the lecture introduces **persona**. A persona is not decoration. It is a way to stop designing for "everyone" and instead design for a specific target user.

Analysis and Design 2025-26

Then comes the most useful line for you:
focus on goals, not the means.
In other words, do not begin with "I want a map" or "I want a dashboard." Begin with:

- what question the user needs answered
- what they are trying to monitor or understand
- what decision the visual should support Analysis and Design 2025-26

Another core idea is **context**. A number by itself means almost nothing. The lecture says measures need context such as:

- timeframe
- benchmarks
- averages
- targets
- historical comparison
- dispersion like min/max or quartiles

Then the lecture gives you a very strong framework: **What – Why – How**.

- **What** = what data is being shown
- **Why** = why the visualisation is needed
- **How** = how the design and interaction should be built Analysis and Design 2025-26

And then the part most students skip: **sketching**. The slides are very clear that sketching is how you build the **skeleton** of the story before adding colour, polish, and aesthetics. Structure first. Style later. Analysis and Design 2025-26

The dashboard design tips in this file are practical and worth memorising:

- group related items
- support meaningful comparisons
- be consistent in encodings
- use legible fonts
- use meaningful colours
- keep supplementary detail available on demand through drill-down/tooltips rather than cluttering the main screen

So the real lesson from this file is:
good visualisation is designed, not assembled.

3. Visualisations for Machine Learning and Explainable AI 2026

This is a different topic. It is not mainly about Tableau coursework design. It is about how visualisation helps with **machine learning understanding and explainability**.

The lecture starts with the point that **accuracy alone is not enough**. If a model is used for loans, jobs, medicine, or public-risk decisions, you need to know:

- why it made a decision
- whether it is biased
- how to challenge a wrong output
- how to improve it Visualisations for Machine Lear…

It then makes another important point:
before understanding the model, understand the data first.
Bias in data leads to bias in the model. Missing or noisy data leads to unreliable outputs. That connects directly with your earlier concern about cleaning and preprocessing.
Visualisations for Machine Lear…

The exploratory data analysis part is actually straightforward. The lecture says EDA helps you:

- identify important variables
- spot inconsistencies
- deal with missing data
- frame meaningful questions
- choose good presentation methods
- generate insights Visualisations for Machine Lear…

It then gives a very practical chart-selection example using Titanic data. That part is useful even outside ML:

- comparison across categories → bar chart
- time change → line chart
- distribution → histogram/box plot/density
- relationship/correlation → scatter/heatmap
- part-to-whole → pie or stacked bar, depending on context
Visualisations for Machine Lear…

The model evaluation part focuses on the **confusion matrix**:

- TP

- FP
- FN
- TN

  And then accuracy, precision, recall, and F1. The most important idea is not just memorising formulas. It is understanding that **different errors matter differently depending on the domain**. The lecture gives the cancer example: false negatives are more dangerous than false positives there.

Then comes **feature importance** and **SHAP**. SHAP is presented as a game-theory-based way to explain how features contribute to a model prediction. The lecture highlights three SHAP plot types:

- **summary plot** for global feature impact
- **force plot** for local reasoning
- **decision plot** for contribution paths

The overall takeaway is that explainability is about **trust, fairness, accountability, and debugging**, not just pretty charts.

What you should understand from all three together

These files are connected more than they look.

The sequence is basically this:

First, understand the user and the question.
Then, choose the right form: dashboard, infographic, or analytic view.
Then, provide context and comparisons.
Then, design interaction and presentation carefully.
Then, if the topic is ML, go beyond "model accuracy" and use visualisation to inspect errors, feature influence, and explainability.

What matters most for you right now

For your immediate lab survival, the most useful parts are:

From **Dashboards and Infographics**:

- dashboard vs infographic difference
- single-screen thinking
- avoid clutter
- use visuals for purpose, not decoration

From **Analysis and Design**:

- persona
- requirements analysis

- context
- What–Why–How
- sketch before building

From **ML/XAI**:

- question first
- pick chart by task
- confusion matrix logic
- precision vs recall tradeoff
- SHAP basics

Announcements / important admin details I found

There is one clear admin note in the ML/XAI lecture: **next week has no lecture or lab**, and **marker feedback on the proposal is upcoming, with dates to be checked on Brightspace**.

The Analysis and Design deck also signals that the following topic after it is **Visualisations for Machine Learning and Explainable AI**, which matches the other uploaded lecture. Analysis and Design 2025-26

Brutal bottom line

Do not reduce this module to "learning Tableau." That is too shallow.

This module is training you to think like this:

Who is the user? What do they need? What question matters? What context is needed? What form fits the task? How do I make it readable? How do I make it interactive only when necessary?

That is the standard.

Send me your **lab sheet or coursework brief next**, and I'll convert all of this into a direct step-by-step action plan for what you should actually do.

Sources

do you remember the old notes from lecture 1 to 5

Thought for 20s

Yes.

In this chat, I still have the lecture set from **1 to 5** in context:

1. **Introduction to Data Visualisation**
2. **Encoding for Perception**
3. **Representation**
4. **Presentation**
5. **Interaction**

And now you also added the later design/dashboard and ML/XAI notes.

So I can still use the earlier notes and connect them with the new ones.
If you want, I'll now give you a **single merged revision guide for Lectures 1–5** in the exact order you should understand them before lab.